

Chromatographic Resolution, R

A certain gas chromatogram showed an air peak, which was a marker for t_m , at 30.0 s and two poorly resolved peaks at 70.0 s and 90.0 s, respectively. All times were measured from injection. The two poorly resolved peaks had base widths of 13.0 s and 15.0 s, respectively.

Calculate R for the two peaks and estimate the % resolution using the normal distribution table.

0z0	y	Area	0z0	y	Area
1.4	0.149 7	0.419 2	2.8	0.007 9	0.497 4
1.5	0.129 5	0.433 2	2.9	0.006 0	0.498 1
1.6	0.110 9	0.445 2	3.0	0.004 4	0.498 650
1.7	0.094 1	0.455 4	3.1	0.003 3	0.499 032
1.8	0.079 0	0.464 1	3.2	0.002 4	0.499 313
1.9	0.065 6	0.471 3	3.3	0.001 7	0.499 517

The resolution expression

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

- Expression can be generated from few algebraic manipulations/estimations starting with $R = \Delta t / w_{av}$
- Use this expression to predict effect of changes to α , k , and N on resolution.
- Do not use this expression to calculate R (use $\Delta t / w_{av}$)

Factors Affecting Resolution

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

Optimum k:

$k = 0 \Rightarrow$ unretained

$$\frac{k_2}{k_2 + 1} = 0$$

$$R = 0$$

$k = \infty \Rightarrow$ long retention

$$\frac{k_2}{k_2 + 1} = 1$$

$$R = R_{\max}$$

$k = 1 \Rightarrow t_r = 2 t_m$

$$\frac{k_2}{k_2 + 1} = 0.5$$

$$R = \frac{R_{\max}}{2}$$

$k = 10 \Rightarrow t_r = 11 t_m$

$$\frac{k_2}{k_2 + 1} \sim 0.9$$

$$R = 0.9 R_{\max}$$

Conclusions: $1 \leq k \leq 10$

$k \sim 10$ is better than $k \sim 1$

$k > 10$ does not significantly improve R

Factors Affecting Resolution

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

Optimum α :

$$\alpha = 1 \quad \Rightarrow \text{peaks co-elute} \Rightarrow \frac{\alpha - 1}{\alpha} = 0 \quad R = 0$$

$\alpha - 1 > 0$... make α as different from 1 as possible

In GC $\Rightarrow$ change the stationary phase

In LC $\Rightarrow$ change the either or both stationary or/and mobile phase

Factors Affecting Resolution

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

Optimum N:

Make N as large as needed.

$$N = \frac{L}{H}$$

$N \propto L$... column length

$$R \propto \sqrt{L}$$

$$R \propto \frac{1}{\sqrt{H}} \quad \text{(more soon about how to decrease H without changing length of the column)}$$

Two peaks in a chromatogram have the following base widths 0.4 and 0.6 minutes. What is the minimum peak separation for complete resolution?

The resolution between two poorly resolved peaks was $R = 0.5$. Would increasing the column length by a factor of 4 lead to complete resolution?

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

Consider two 25 cm x 2.6 mm id columns on which the same solute is analyzed. On column A, we measure a retention time of 416.4 s and a peak width = 17.3 s. The corresponding values on column B are $t_r = 481.2$ s and $w = 22.4$ s. Which column is more efficient?

In chromatography, _____ plate height, H ,
gives narrower peaks

A. smaller

B. larger